

Department of
Computer Science
Faculty of Exact Sciences
Bar-Ilan University

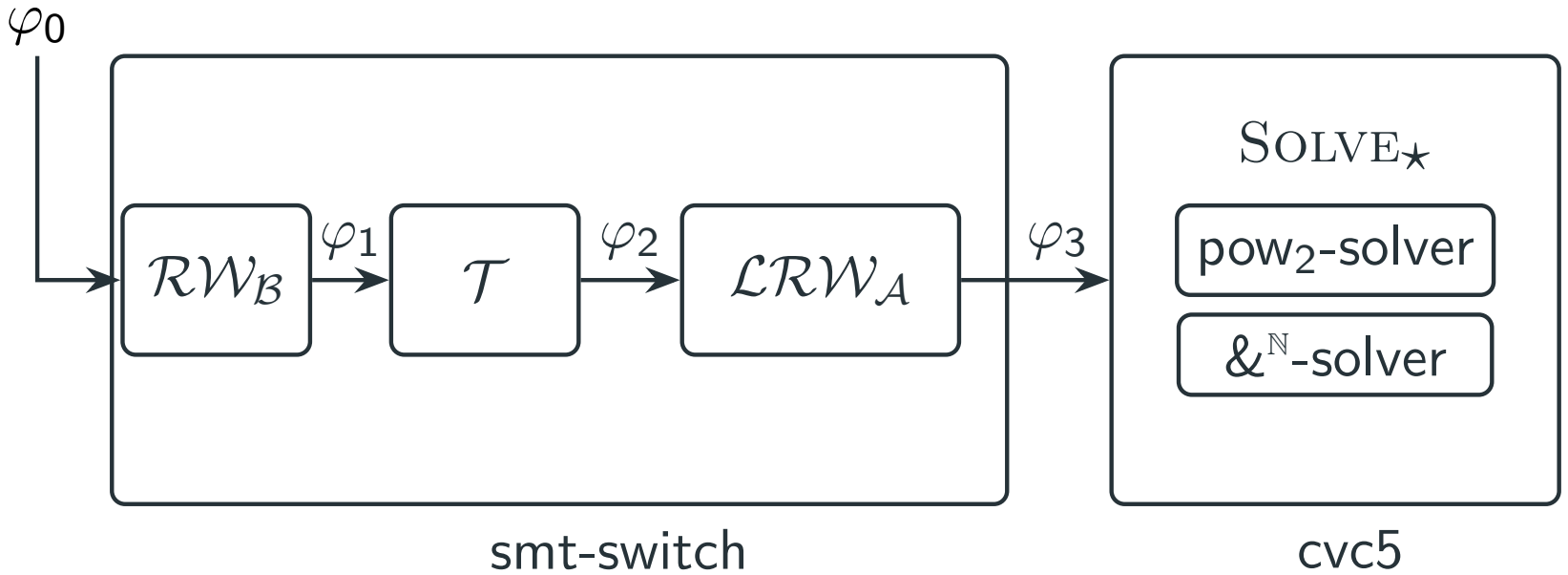
# Parametric Bit Vector

**Zvika Berger**

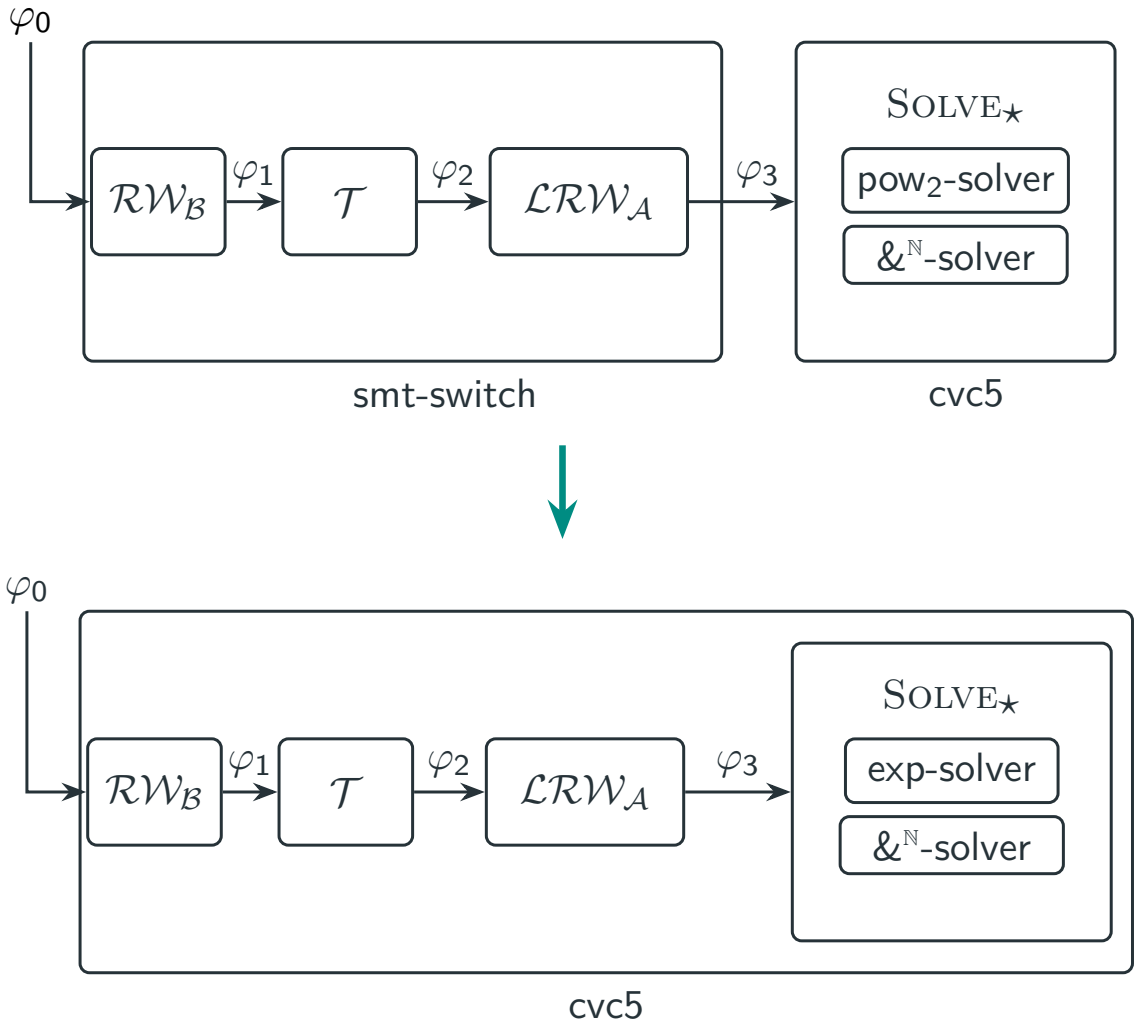
cvc5 summit 2026



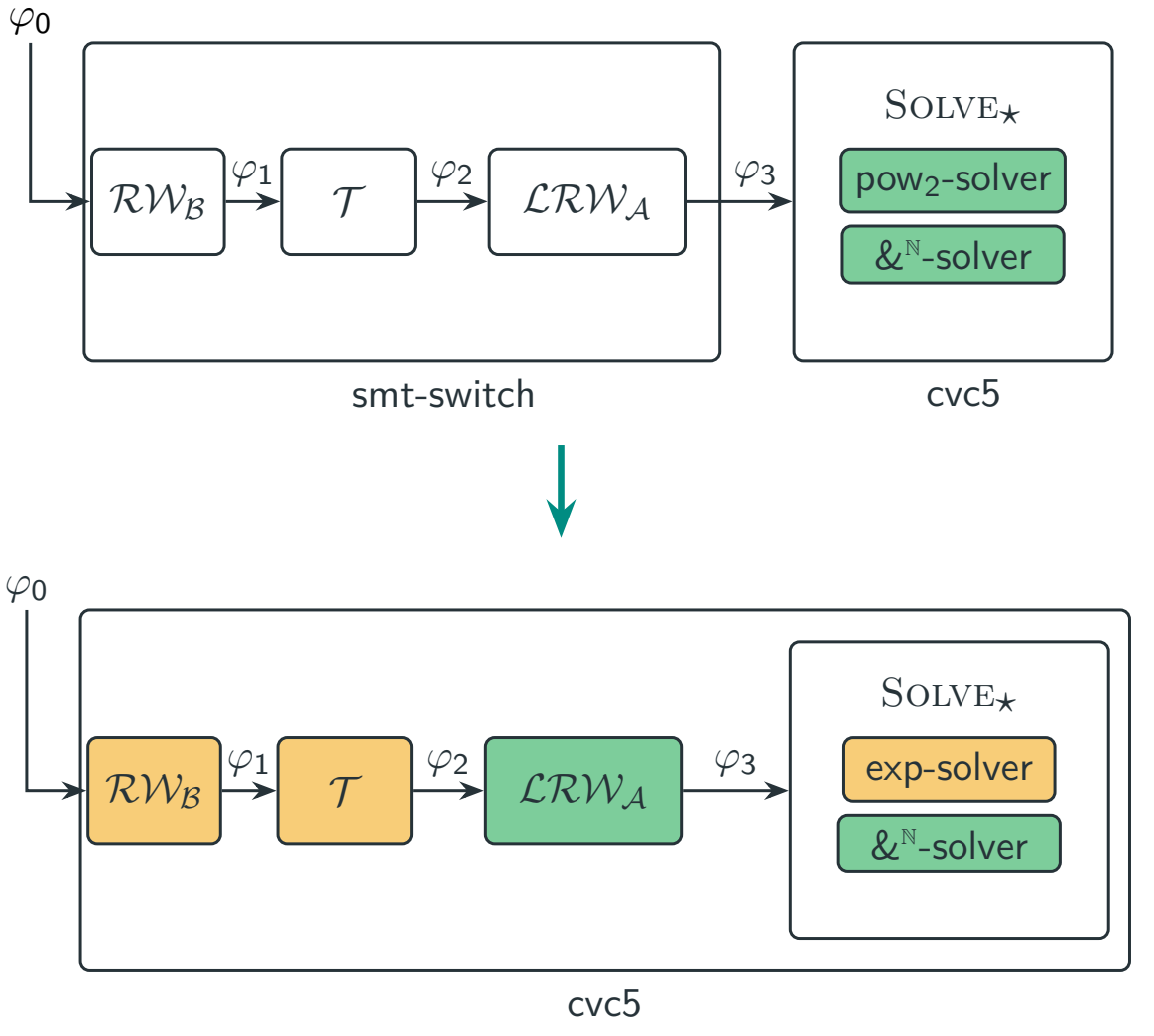
# SAT'25 Architecture



# Integration to cvc5



# Integration Status




Merged to cvc5 main  Implemented in fork

# New Contributions (Paper Coming Soon)

## Improved parametric bit-vector solving, over:

**SAT'25**[1]
our previous work, bit-precise PBV reasoning.

**CAV'26**[2]
multi-width PBV equivalence solver.

**OOPSLA'26**[3]
linear-bitwise PBV solver.

## Improved exponential solving, over:

**IJCAR'24**[4]
SMT for exponential integer arithmetic.

## New sets of benchmarks for parametric bit-vectors.



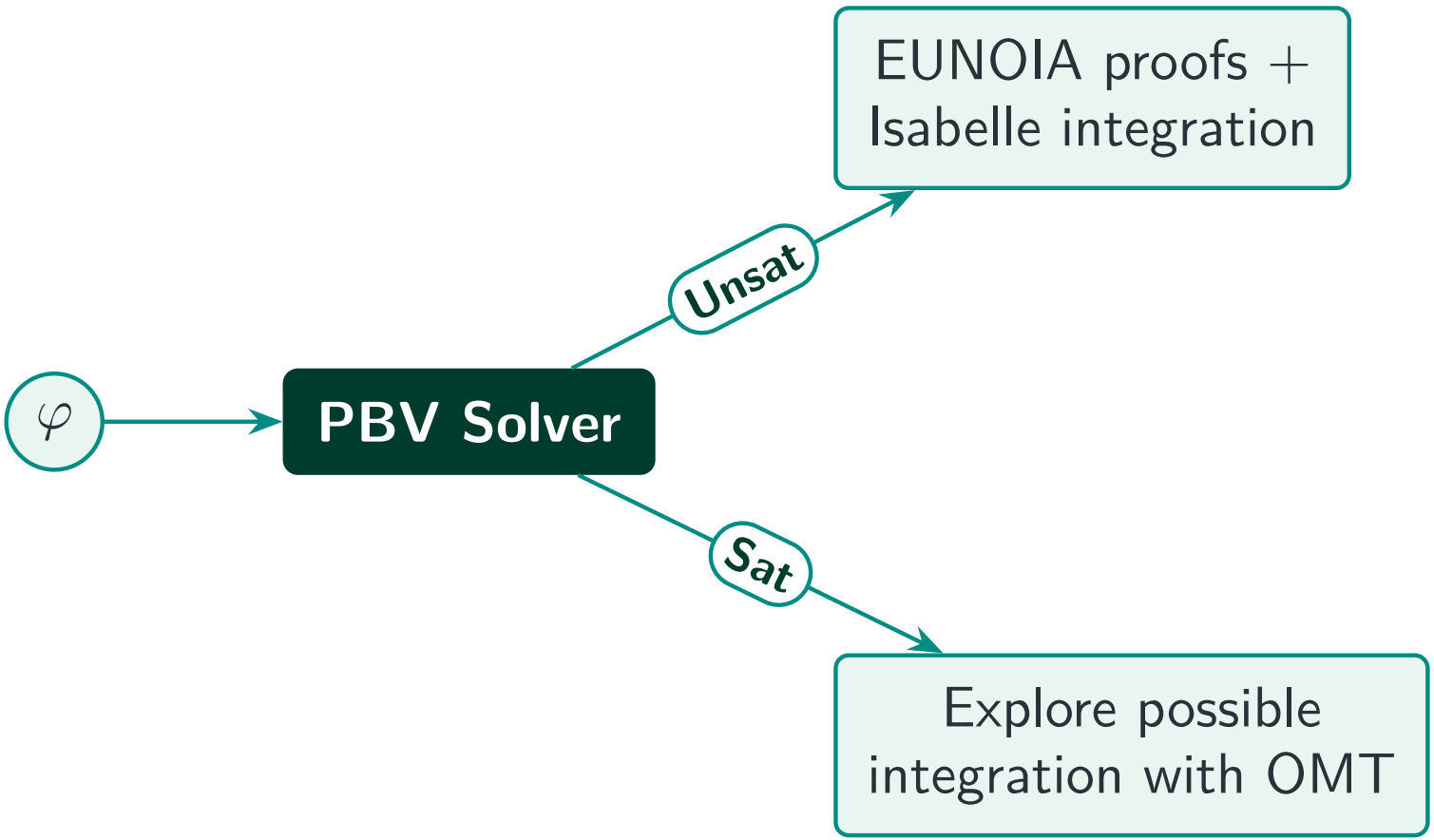

[1] Berger et al. *Bit-precise reasoning with parametric bit-vectors.* SAT'25.

[2] Rinaldi et al. *A multi-width parametric bitvector equivalence solver.* CAV'26.

[3] Bhat et al. *Sound and complete solving for multi-width parametric bitvectors via principled reductions.* OOPSLA'26.

[4] Frohn and Giesl. *Satisfiability modulo exponential integer arithmetic.* IJCAR'24.

# Future Work


EUNOIA proofs +
Isabelle integration
Unsat
$\varphi$
PBV Solver
Sat
Explore possible
integration with OMT